

Philosophy of Science
Prof. James W. McAllister

Lecture 12

Review and Question Period

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Today's programme

- Preparation for final exam
 - Including practice exam questions
 - Information first presented in lecture 7
- Bird's-eye view of the course
 - Lecture-by-lecture overview
- Any further business

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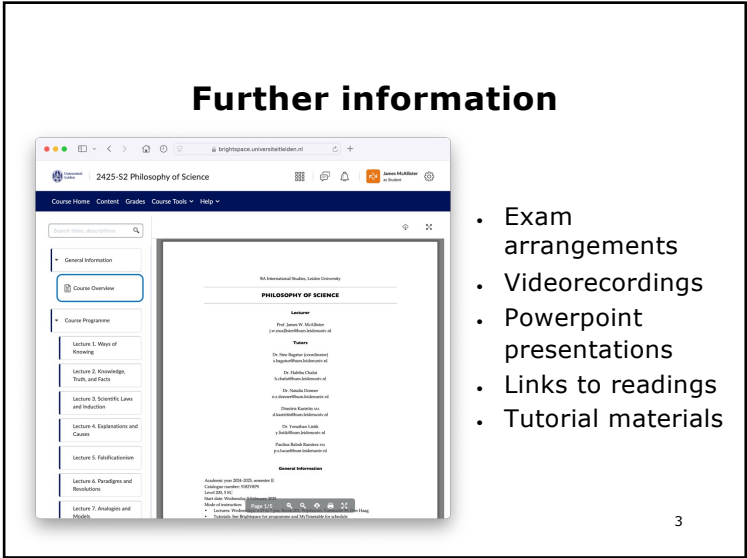
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Final exam preview

- Written exam, two hours
 - Part 1: twenty multiple-choice questions
 - Part 2: two open essay questions—max. 300 words per question
 - Parts 1 and 2 have equal weight
 - Closed book
- Platform Ans on Chromebooks
 - See MyTimetable for date, time, place
- We now review some practice questions
 - Three multiple-choice, two open essay

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Multiple-choice question 1

Which of the following best characterises the nomothetic approach?

- A. It is a model of explanation proposed by Carl G. Hempel in the 1960s
- B. It is typical of the natural sciences
- C. It cannot be applied to human and social phenomena
- D. It is incompatible with induction

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Multiple-choice question 2

The correspondence theory of truth says that a statement P is true if and only if...

- A. We are justified in believing that P
- B. We know P to be true
- C. P corresponds with the facts
- D. We have convincing empirical evidence for P

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Multiple-choice question 3

What does it mean to say that induction is ampliative?

- A. Induction rests on ample empirical evidence
- B. Inductive inferences give researchers secure grounds for their conclusions
- C. Induction goes from the general to the particular
- D. Inductive inferences go beyond their premises

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Answers and references

- . Question 1: B
 - . Lecture 1, sheet 27
- . Question 2: C
 - . Lecture 2, sheet 19
- . Question 3: D
 - . Lecture 3, sheet 23

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Open essay question 1

- A. According to the JTB account of knowledge, under which conditions may we assert "Andrea knows that Bob was born in Paris"? Explain the meaning of any philosophical terms that you use. (5 marks)
- B. Assume that Andrea knows that Bob was born in Paris. Then, explain how it is possible for Chris to believe that Bob was not born in Paris, but not possible for Chris to know that Bob was not born in Paris. (3 marks)
- C. Is it possible to have a justified belief that P where P is a proposition that is not true? Explain your reasoning. (2 marks)

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Open essay question 2

- A. What are the requirements for a successful analogy in science? Explain the meaning of any technical terms that you use. (5 marks)
- B. Using the framework that you have just laid out, develop an analogy between an electronic computer and a human brain. (3 marks)
- C. How can analogy lead to new discoveries in science? Give an example as part of your answer. (2 marks)

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Components of good essays

- . Introduction to the philosophical context
- . Exposition and explanation of relevant philosophical concepts and theories
- . Reasoning and arguments establishing pertinent conclusions
- . Examples and illustrations
- . Discussion of further implications

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The course: bird's-eye view 1

I. THE BASICS

- 1. Ways of knowing
- 2. Knowledge, truth, and facts

II. SOME SCIENTIFIC DEVICES

- 3. Scientific laws and induction
- 4. Explanations and causes

III. MODELS OF SCIENTIFIC PRACTICE

- 5. Falsificationism
- 6. Paradigms and revolutions

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Bird-eye's view 2

IV. METHODOLOGICAL TOOLBOX

- 7. Analogies and models
- 8. Quantitative and qualitative approaches

V. RATIONALITY AND CULTURE

- 9. Rationality
- 10. Interpretation and understanding
- 11. Absolutism and relativism
- 12. Review and question period ← *Today*

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1. Ways of knowing

- Three groups of sciences
 - Natural, human, and social sciences
 - Variety within unity
 - Characteristics, strengths, and blind spots
- Distinction between two approaches to describe the world
 - Nomothetic and idiographic approaches
 - Key to interpreting the knowledge that each discipline can produce

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2. Knowledge, truth, and facts

- JTB account of propositional knowledge
 - *A* knows that *P* if and only if: *P* is true; *A* believes that *P*; *A* is justified in believing that *P*
- Correspondence theory of truth
 - A proposition *P* is true if and only if there is a fact corresponding to it
- Facts: what make propositions true
 - Facts—and therefore truths—do not change

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3. Scientific laws and induction

- Nomothetic approach
 - Typical of the natural sciences, used also in human and social sciences
- Concept of scientific law
 - Four examples in human and social sciences
- Induction as a technique for...
 - detecting regularities and formulating laws
- David Hume's problem of induction
 - And a pragmatic justification of inductivism

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4. Explanations and causes

- Explanation and...
 - knowledge; truth; prediction
- Explanation as deduction
 - Carl G. Hempel: deductive-nomological, inductive-statistical models
- Criticism of these models
 - Failure to track causal relations
- New causal approaches to explanation
 - Wesley C. Salmon: causal-mechanical model

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5. Falsificationism

- Karl R. Popper's principles
 - Asymmetry of confirmation and refutation
 - Rejection of induction
 - Distinction discovery / justification
 - Demarcation science / non-science
- Four lines of criticism of falsificationism
 - Refutation not straightforward; lack of accord with scientific practice; no effort to model confirmation; some hypotheses not falsifiable

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6. Paradigms and revolutions

- Thomas S. Kuhn
 - *The Structure of Scientific Revolutions*
 - First edition 1962, second edition 1970
- Key concepts
 - Theory ladenness of observation
 - Paradigm
 - Scientific revolution
 - Incommensurability: semantic, observational, methodological

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7. Methodology of analogies and models

- What is analogy?
 - Four components: source domain, target domain, mapping, relations
 - Heuristics of analogy: neutral analogy
- Historical examples
 - Water / light, organism / society
- What is a model?
 - Models and idealisation
 - Models and analogy

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8. Quantitative and qualitative approaches

- Comparison on two dimensions
 - Hypothesis formulation vs. test
 - Forward vs. reverse causal inference
- Case-study research
 - Paradigmatic example of qualitative method
- Internal and external validity
 - Trade-off between warrant and generalisability of findings

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9. Rationality

- Rationality: property of an action of being optimal under a set of constraints
- Two uses of models of rationality
 - Descriptive and normative
- Two models of rationality
 - Rational choice theory
 - Bounded rationality theory
- Coda: rationality and emotion

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10. Interpretation and understanding

- Wilhelm Dilthey's distinction
 - Natural and mental phenomena
- Relationality of mental phenomena
 - How we individuate them in experience
 - Role of *verstehen*
- Further implications of relationality
 - Purposiveness, meaningfulness
 - How to do research in human sciences

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11. Absolutism and relativism

- What are absolutism and relativism?
 - Claims about how entities possess a given property *F*—absolutely or relatively
- Relativism about truth
 - Self-undermining thesis, according to analytic epistemology
- Relativism about rationality
 - Fruitful stance in history of science and cultural anthropology

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Any further questions?

- I will be delighted to respond!

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Any other business?

- Final chance in this session to...
 - pose any remaining questions
 - raise any further issues

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And lastly...

A big thank you,

good luck with the final exam,

and good-bye!

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